

Momin N. Siddiqui

Georgia Institute of Technology

Advisor: Chris MacLellan

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Research Statement

My research explores how human-centered AI can support learning, reasoning, and writing. I design intelligent systems that scaffold cognition and augment student thinking to create accessible and reflective learning experiences.

Education

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|-------------|---|
| 2023 – 2025 | Georgia Institute of Technology, Atlanta, GA, USA
M.S. Computer Science (Specialization: HCI), GPA: 4.0/4.0 |
| 2019 – 2023 | Jamia Millia Islamia University, Delhi, India
B.Tech. Computer Engineering, GPA: 9.41/10 |

Research Experience

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|----------------|---|
| 2023 – Present | Georgia Tech, Teachable AI Lab, Graduate Research Assistant
Developing methods and frameworks for Intelligent Tutoring Systems focusing on adaptive scaffolding and agent evaluation. |
| 2024 | Stanford, Institute for Human-Centered AI, Research Assistant
Designed and evaluated human-AI interaction methods for collaborative writing tools to support cognitive processes and augment student abilities. |
| 2022 – 2023 | IIT Delhi, Human-Machine Interaction Lab, Research Intern
Developed computational methods using neuropsychological tests to assess student attention during online learning. |

Selected Publications

- [1] **Momin N. Siddiqui**, Roy Pea, Hari Subramonyam. “Script&Shift: A Layered Interface Paradigm for Integrating Content Development and Rhetorical Strategy with LLM Writing Assistants.” *Proceedings of the CHI Conference on Human Factors in Computing Systems* (CHI 2025). 🏆 **Honorable Mention (Top 5%)**.
- [2] **Momin N. Siddiqui**, Vryan Feliciano, Roy Pea, Hari Subramonyam. “AI in the Writing Process: How Purposeful AI Support Fosters Student Writing.” *Proceedings of the International Conference on Artificial Intelligence in Education* (AIED 2025).
- [3] Daniel Weitekamp*, **Momin N. Siddiqui***, Chris MacLellan. “TutorGym: A Testbed for Evaluating AI Agents as Tutors and Students.” *Proceedings of the International Conference on Artificial Intelligence in Education* (AIED 2025).

- [4] **Momin N. Siddiqui**, Adit Gupta, Jennifer Reddig, Chris MacLellan. “HTN-Based Tutors: A New Intelligent Tutoring Framework Based on Hierarchical Task Networks.” *Proceedings of the Eleventh ACM Conference on Learning at Scale (L@S 2024)*.
- [5] Saumya Yadav, **Momin N. Siddiqui**, Yash Vats, Jainendra Shukla. “EngageME: Exploring Neuropsychological Tests for Assessing Attention in Online Learning.” *Proceedings of the International Conference on Artificial Intelligence in Education (AIED 2024)*.

(* denotes equal contribution)

Awards

2024 Foley Scholar Award

Invited Talks

2024 **Contextualizing Writing Education for LLM Instigated Paradigm Shift**
Georgia Tech IPaT Lunch Lecture, *Guest Lecture*

Leadership and Outreach

2023 – 2024 **Tikkun Olam Makers**, *Project Lead*
Led a team of 10 undergraduate students to develop an AAC device for a client with Rhett syndrome.

2019 – 2023 **Tamana Foundation**, *Special Educator*
Assisted neurodivergent children with education and rehabilitation needs.

Technical Skills

Languages Python, JavaScript, C++, C#, Bash, Kotlin, HTML/CSS, R, SQL

Libraries React.js, Node.js, PyTorch, Tensorflow, Docker, Kubernetes, Heroku, Pandas, Scikit-Learn, Numpy, Unity, React Native

Experience Human-Computer Interaction, NLP, Deep Learning, Knowledge-Based AI, Cognitive Science, Learning Sciences, Multimodal User Interfaces, Web Development, Android Development, Game Development, Backend Programming, User Studies, Data Collection, Annotation, Crowdsourcing